

Monday

1. HW 7.4 Due
 - A. Questions?
2. Worksheet 7.5.2
3. Practice Problems
4. 7.8 Improper Integrals

Wednesday

1. Review for test
2. Finish 7.8.

Friday

1. Test 2 (Start at 1:30)

$$\int \frac{-21x+1}{(5x-1)(x+3)} dx \quad \left[\frac{-21x+1}{(5x-1)(x+3)} = \frac{A}{5x-1} + \frac{B}{x+3} \right] (5x-1)(x+3)$$

$$\begin{aligned} -21x+1 &= A(x+3) + B(5x-1) \\ &= Ax + \underline{3A} - 5Bx - \underline{B} \end{aligned}$$

$$\begin{aligned} \underline{x=-3} \\ 64 &= B(-16) \end{aligned}$$

$$-4 = B$$

$$\begin{aligned} \underline{\text{constant}} \\ 1 &= 3A - B \\ 1 &= 3A + 4 \\ -3 &= 3A \\ -1 &= A \end{aligned}$$

$$\int \frac{-1}{5x-1} + \frac{-4}{x+3} dx = \boxed{-\frac{1}{5} \ln|5x-1| - 4 \ln|x+3| + C}$$

7.8 Improper Integrals

Recall: We can interpret a definite integral as the area of a region.

So far, all of the definite integrals we've done involved regions with finite height and width.

Improper Integrals occur when either

- Type I: The region has infinite width:

$$\int_{-\infty}^b f(x) dx, \quad \int_a^{\infty} f(x) dx, \quad \text{or} \quad \int_{-\infty}^{\infty} f(x) dx$$

- Type II: The region has infinite height:

$$\int_a^b f(x) dx, \text{ where } f(x) \text{ has a vertical asymptote between } a \text{ and } b.$$

1 Type I: Infinite endpoint(s)

Fact: Let a be any positive number. Then the integral

$$\int_a^{\infty} \frac{1}{x^n} dx \quad \text{is} \quad \begin{cases} \text{infinite} & \text{if } n \leq 1 \\ \text{finite} & \text{if } n > 1 \end{cases}$$

Check it out here: <https://www.desmos.com/calculator/ztmfssfpfb>

These next three examples show why this rule holds true:

1.0.1 Example:

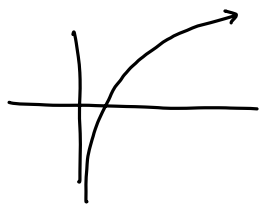
Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{x^3} dx = -\frac{1}{2} x^{-2} \Big|_1^{\infty} = -\frac{1}{2x^2} \Big|_1^{\infty} = \lim_{b \rightarrow \infty} \left[-\frac{1}{2b^2} \right]_1^b = \lim_{b \rightarrow \infty} \left[-\frac{1}{2b^2} + \frac{1}{2} \right] = \frac{1}{2}$$

1.0.2 Example:

Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{x} dx = \ln|x| \Big|_1^{\infty} = \lim_{b \rightarrow \infty} \ln x \Big|_1^{\infty} = \lim_{b \rightarrow \infty} \underbrace{\ln b}_{\infty} - \underbrace{\ln 1}_0 = \infty$$



1.0.3 Example:

Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{\sqrt{x}} dx = \int_1^{\infty} x^{-1/2} dx = 2x^{1/2} \Big|_1^{\infty} = \lim_{b \rightarrow \infty} 2\sqrt{x} \Big|_1^{\infty} = \lim_{b \rightarrow \infty} \underbrace{2\sqrt{b}}_{\infty} - \underbrace{2\sqrt{1}}_2 = \infty$$

1.1 Integrals from $-\infty$ to ∞

For integrals like this, you have to break them into two separate integrals, and do them independently:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx$$

The break-point b can be any real number, so choose anything you want (0 is often convenient).

1.2 Comparisons

What about an integrals such as

$$\int_1^{\infty} \frac{1}{e^n} dx \quad \text{or} \quad \int_3^{\infty} \frac{1}{x^2 + x} dx$$